Now that we have the global eliminator for the free Cartesian category CC[Σ] stated as a global section of any displayed Cartesian category over it, we can use the eliminator to describe a logical relation via category gluing. That is, we need to construct a displayed Cartesian category over CC[Σ] such that the existence of a section carries our proof

A prototypical example of a metatheoretic proof by logical relation is demonstrating the canonicity property

↳ canonicity says that closed terms of any type reduce to canonical forms.
In essence, canonicity states that the equational theory of the language is sufficient to specify the behavior of an interpreter

How do we build a displayed Cartesian category?
Well, what is the precise definition of a displayed cartesian category?

Recall that a Cartesian category is one with a terminal object and all binary products. Another way to capture those universal properties is via representability of certain presheaves

A terminal object represents the terminal presheaf. $\text{よ} \cong 1_{psh}$

A product of A and B represents よA × よB

$$\text{よ}(A \times B) \cong \text{よ}A \times \text{よ}B$$

Correspondingly, we can give a description of displayed universal properties with a straightforward adaption of the ordinary universal properties to the displayed setting

Displayed Presheaf

A displayed presheaf P on a displayed category $\begin{smallmatrix} D \\ \downarrow \\ C \end{smallmatrix}$ over a

Presheaf Q on C is a displayed functor from D to a displayed category of sets, $\begin{smallmatrix} \text{Depset} \\ \downarrow \\ \text{Set} \end{smallmatrix}$

$$\begin{tikzcd} D^{op} \arrow[r, "P"] \arrow[d] & \text{Depset} \arrow[d] \\ C^{op} \arrow[r, "Q"] & \text{Set} \end{tikzcd}$$

We also give an alternative characterization of displayed presheaves later

We may then define the representable displayed presheaves as those describing sets of displayed morphisms, just as the ordinary representable presheaves describe sets of morphisms into a fixed object

That is, for $d : D_c$

$\text{よ}_{Disp}\, d$ is displayed presheaf on D, displayed over the presheaf よc on C

We may then develop the appropriate notions of natural transformation and isomorphism between displayed presheaves

Representability (a displayed presheaf $\begin{smallmatrix} P \\ \downarrow \\ Q \end{smallmatrix}$ then can be characterized in the same way that representability is

i.e. $\begin{smallmatrix} P \\ \downarrow \\ Q \end{smallmatrix}$ representable if there is a choice of $c : C$ and $d : D_c$ such that $Q \cong \text{よ}c$ and $P \cong \text{よ}_{Disp}\, d$

Similarly, displayed terminals represent the terminal displayed presheaf and displayed products of objects of $A : D_x, B : D_y$ represent

$$\begin{smallmatrix} \text{よ}_{Disp} A \times_{Disp} \text{よ}_{Disp} B \\ \downarrow \\ \text{よ}x \times \text{よ}y \end{smallmatrix}$$

Actually constructing such a displayed Cartesian category for a canonicity proof.

To do so, we will use a reindexing construction on displayed categories

Reindexing

If $\begin{smallmatrix} D \\ \downarrow \\ C \end{smallmatrix}$ is a displayed category, and we have a $F : B \to C$ then we can reindex F to be a displayed category over B

↳ Essentially, we take the displayed category D, the image of the functor F.

$$(\text{reindex}_F\, D)_c := D_{F(c)}$$

↳ should we write $F^* D$?

Lastly, we can give a slightly different interface for proving displayed cartesianess

Rather than directly constructing a representing object for $\text{よ}^D A^D \times^D \text{よ}^D B^D$

we can apply the following lemma,

Lemma C Cartesian

$\begin{smallmatrix} D \\ \downarrow \\ C \end{smallmatrix}$ has cartesian lifts along projections π_1 and π_2

For all $c : C$, the fiber category D_c is Cartesian.

Then D is displayed cartesian.

We refer to the precondition to this lemma as stating that D is vertically Cartesian

Throughout this work, displayed usually refers to a structure throughout the entire displayed category — its what you get from having the total category have the universal structure and $\pi_1 : \int D \to C$ preserve the structure strictly.

Vertical means fiberwise/local

Vertical cartesianess is preserved by reindexing. So our canonicity proof is thus

$\begin{smallmatrix} \text{Set}^D \\ \downarrow \\ \text{Set} \end{smallmatrix}$ is a vertical cartesian category

The global sections functor $CC[\Sigma][1, -] = C(-)$ maps $CC[\Sigma]$ into Set.

We may then reindex Set^D to be a vertical cartesian category over CC[Σ], then apply the lemma to show that $\text{reindex}_{C(-)}\, \text{Set}^D$ is displayed cartesian

↳ Also need a lemma that reindexing preserves the existence of cartesian lifts

The remaining part of the proof is left in showing that $\text{reindex}_{C(-)}\, \text{Set}^D$ interprets all of the generators of Σ

If Σ has, say, base types for Booleans and natural numbers, we must chose interpretations for each.

For canonicity we map each type to a dependent set over its closed terms, and in particular we map to the set of proofs that a closed term is in canonical form.

For each Bool and ℕ we can map a semantic value into a syntactic one

QuoteBool b := if b then tt else ff

Quoteℕ 0 := z
Quoteℕ (suc n) := Quoteℕ n ★ s
or
s (Quoteℕ n)

And the canonical forms of $\cdot \vdash b : \text{Bool}$ or $\cdot \vdash n : \mathbb{N}$ are proofs that b or n are each in the image of the quote maps

↳ i.e. the boolean is either true or false; the natural is a numeral

Then we must show that the interpretations of Σ {tt, ff, z, s} respect these canonical forms, which is easy

Finishing the interpretation of Σ in $\text{reindex}_{C(-)}\, \text{Set}^D$, we then get to invoke the eliminator to get a global section

$$\begin{tikzcd} \text{reindex}_{C(-)}\, \text{Set}^D \arrow[d] \\ CC[\Sigma] \arrow[u, bend left, "\text{elim}"] \end{tikzcd}$$

The section given by elim describes for all syntactic types in CC[Σ] the construction of a canonical form inductively.

It is exactly the same as we would expect had we constructed the logical relation by hand.

Intuitively we would expect canonical forms of Bool × ℕ to a pair of canonical forms for a Bool and a ℕ

The section given by the eliminator strictly preserves products. That is,

elim acting on a closed term of type Bool × Nat will return the displayed product of the motive for Booleans and the motive for naturals

What is the displayed product in $\text{reindex}_{C(-)}\, \text{Set}^D$?

$$\iota(\text{Bool}) \times^D \iota(\mathbb{N}) \quad \text{in } \text{Set}^D$$

$$\triangleq \pi_1^*(\iota(\text{Bool})) \times^V \pi_2^*(\iota(\mathbb{N}))$$

and cartesian lifting here is just composition, vertical binary product is just the product in the image

So, that's all to say, for $e : \text{Bool} \times \mathbb{N}$

$$\text{elim } e : \text{isCanonicalBool}(\pi_1(e)) \times \text{isCanonical}\mathbb{N}(\pi_2(e))$$

which is precisely what we would expect from the manual construction. We let the category theory do some of the bookkeeping for us, as in the manual proof we really had no choice in how we were to define the logical relation for product types, it was always going to be as products of canonical forms

The same principles hold true for adding additional connectives. Coproducts are straightforward, although one must be careful in encoding function types